KneeXAI: Explainable AI System for Osteoarthritis Severity Classification

This system is developed for educational and research purposes only. It is not intended for clinical diagnosis or medical decision-making.

Inference

Model

ResNet-50

Use FastAPI endpoint

FastAPI URL

http://127.0.0.1:8000

Explainability

Show XAI visualization

XAI methods

Grad-CAM ×
Grad-CAM++ ×
Eigen-CAM ×
Integrated Gradi… ×
SmoothGrad ×

Overlay strength

0.45

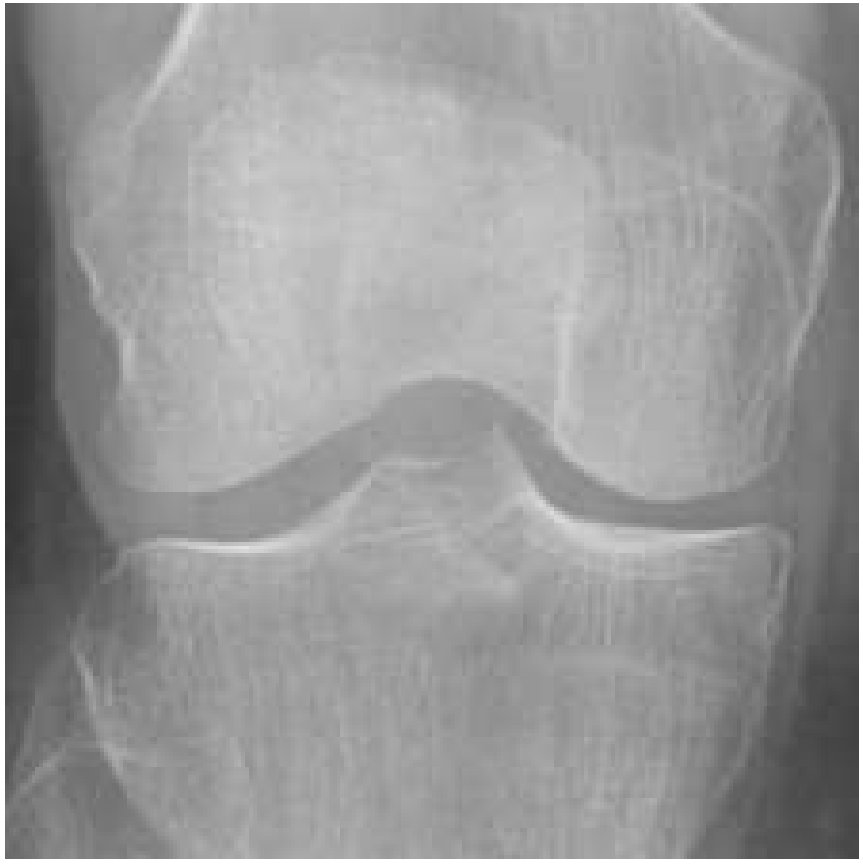
0.10 0.80

Upload a knee X-ray image

Drag and drop file here
Limit 200MB per file • PNG, JPG, JPEG, BMP, WEBP

Browse files

kl-1-1.png 22.8KB



Uploaded image

Predicted class: Grade 1

Meaning: Doubtful osteoarthritis with possible osteophytes or minimal narrowing.

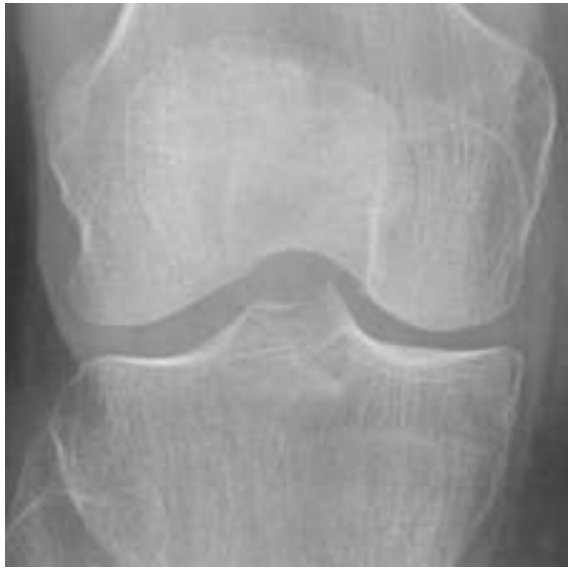
Confidence
58.7%

Latency
32.2 ms

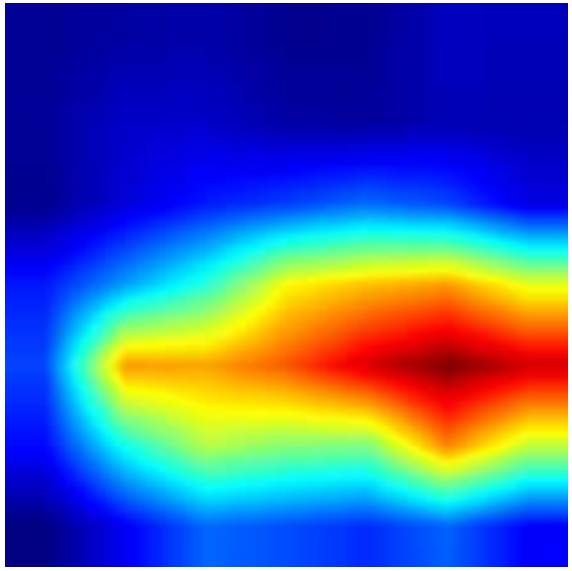


Explainable AI visualizations

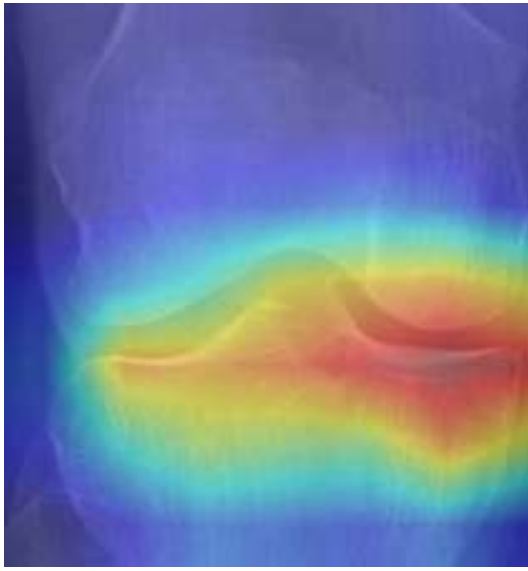
Grad-CAM | Grad-CAM++ | Eigen-CAM | Integrated Gradients | SmoothGrad



Original X-ray

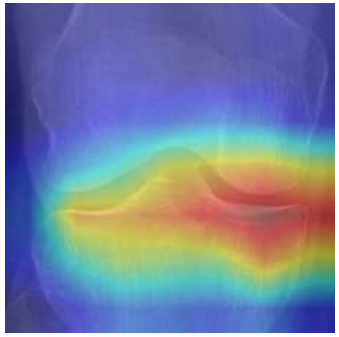


Grad-CAM heatmap

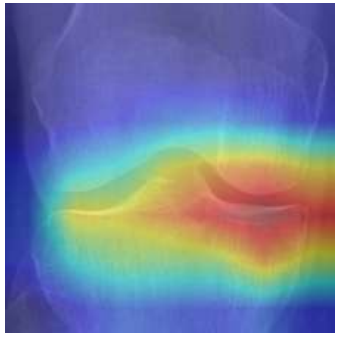


Grad-CAM overlay

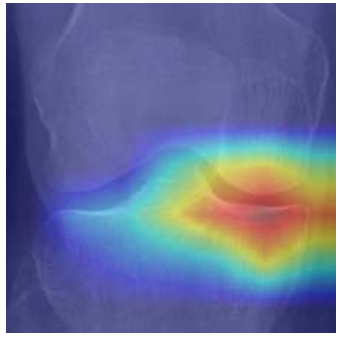
Overlay comparison



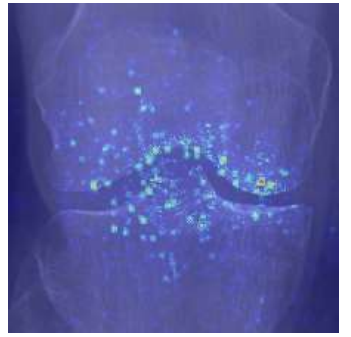
Grad-CAM



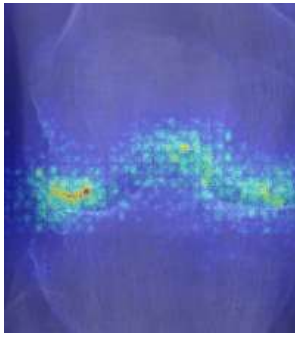
Grad-CAM++



Eigen-CAM



Integrated Gradients



SmoothGrad

Quantitative XAI comparison

Comparison	Pearson correlation	Top 20% IoU	Agreement score
Grad-CAM vs Grad-CAM++	0.998	0.922	0.960
Grad-CAM vs Eigen-CAM	0.931	0.739	0.835
Grad-CAM vs Integrated Gradients	0.402	0.284	0.343
Grad-CAM vs SmoothGrad	0.548	0.325	0.436
Grad-CAM++ vs Eigen-CAM	0.941	0.801	0.871
Grad-CAM++ vs Integrated Gradients	0.407	0.287	0.347
Grad-CAM++ vs SmoothGrad	0.562	0.331	0.447
Eigen-CAM vs Integrated Gradients	0.355	0.269	0.312
Eigen-CAM vs SmoothGrad	0.473	0.299	0.386
Integrated Gradients vs SmoothGrad	0.420	0.317	0.369

Method	Focus area ratio
Grad-CAM	0.
Grad-CAM++	0.
Eigen-CAM	0.
Integrated Gradients	0.
SmoothGrad	0.

Average XAI Agreement Score

0.531

The XAI methods show moderate agreement, suggesting some shared focus with method-specific differences.

Heatmap similarity matrix

	Grad-CAM	Grad-CAM++	Eigen-CAM	Integrated Gradients	SmoothGrad
Grad-CAM	1.000	0.998	0.931	0.402	0.548
Grad-CAM++	0.998	1.000	0.941	0.407	0.562
Eigen-CAM	0.931	0.941	1.000	0.355	0.473
Integrated Gradients	0.402	0.407	0.355	1.000	0.420
SmoothGrad	0.548	0.562	0.473	0.420	1.000

Deletion Check

Method	Original class confidence	After deletion confidence	Confidence drop	Deleted prediction
Grad-CAM	0.587	0.484	0.102	Grade 1
Grad-CAM++	0.587	0.483	0.104	Grade 1
Eigen-CAM	0.587	0.518	0.069	Grade 1
Integrated Gradients	0.587	0.028	0.558	Grade 3
SmoothGrad	0.587	0.001	0.586	Grade 4

Average confidence drop after deleting top 20% XAI region

0.284

Deletion Check shows a moderate confidence drop, so the highlighted regions contain useful decision evidence.

Stability Check

Method	Perturbation	Pearson correlation	Top 20% IoU	Stability score
Grad-CAM	brightness +5%, contrast +5%	0.999	0.949	0.
Grad-CAM++	brightness +5%, contrast +5%	0.999	0.965	0.
Eigen-CAM	brightness +5%, contrast +5%	0.999	0.970	0.
Integrated Gradients	brightness +5%, contrast +5%	0.922	0.689	0.
SmoothGrad	brightness +5%, contrast +5%	0.859	0.668	0.

Average Stability Score

0.902

Stability Check is high: small brightness and contrast changes keep the explanation mostly consistent.